

Local Identity Records

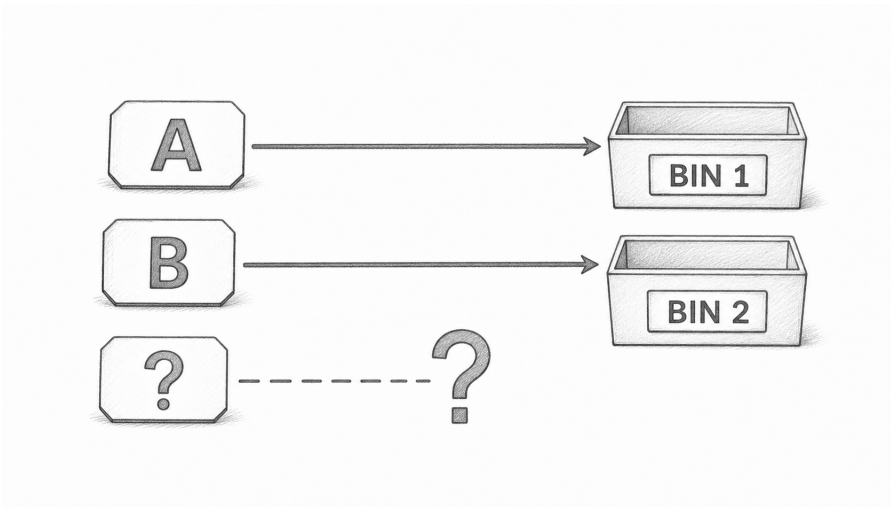
Map copied lesson tokens to bounded paper-bin records

E0 SYNTHETIC RECORD REPLAY — NO RFID OR STORAGE CLAIM

Lesson	049 — component	Time	90–120 minutes
Prerequisites	status, explicit time, fixed records	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no reader, keypad, bus, or medium

Responsibility. Validate a copied UID-shaped observation, map it to one bounded local paper-bin record, and stage enrollment as an explicit transaction. An identifier is a lesson-local token: not a person, credential, secret, proof of possession, authentication result, or security boundary.

1 Follow a token into a local directory

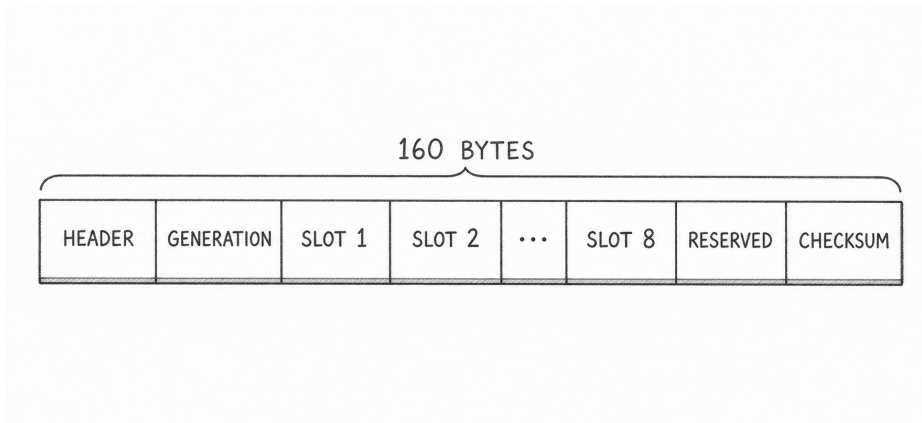


Alternative text: a graphite pencil drawing maps card A by a solid arrow to Bin 1 and card B by a solid arrow to Bin 2. A third card marked with a question mark follows a dashed line to a larger question mark.

LocalIdentityRegistry owns no reader, keypad, pin, endpoint, timer, interrupt, bus, clock, storage transport, supply, or moving hardware. It borrows a live binding array, exactly two 160-byte image slots, and one 160-byte candidate buffer. These caller-owned arrays must outlive the registry and must not move.

Fixed limit	Value	Meaning
identity bytes	10	4–10 significant bytes, then canonical zero padding
local bindings	8	allocation-free compile-time maximum
carousel bins	8	local metadata, not part of identity equality
image slots	2	complete supplied images, never an inferred empty directory
image bytes	160	canonical byte encoding, not a copied C++ struct

2 Read a canonical fixed image



Alternative text: a graphite pencil drawing shows one 160-byte horizontal record divided into Header, Generation, Slot 1, Slot 2, an ellipsis, Slot 8, Reserved, and Checksum regions.

Multibyte integers are unsigned little-endian. Reserved bytes and unused entries are zero. CRC-16/CCITT-FALSE detects malformed fixture bytes; it does not provide authenticity, secrecy, or adversarial integrity.

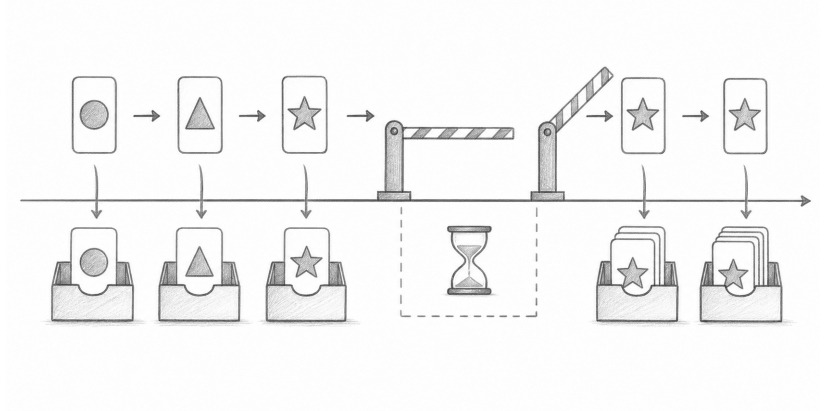
Offset	Bytes	Canonical meaning
0	2	magic 0x4944
2	1	schema version 1
3	1	flags, exactly zero
4	2	encoded length, exactly 160
6	4	nonzero registry configuration ID
10	4	image generation
14	1	binding count, 0–8
15	1	reserved zero
16	128	eight fixed 16-byte binding entries
144	14	reserved zero
158	2	checksum over bytes 0–157

Each used entry stores length at +0, ten zero-padded identity bytes at +1... + 10, bin at +11, nonzero revision at +12... + 13, and an entry checksum at +14... + 15. All-zero and all-0xff identifiers are fixture sentinels and are rejected.

3 Reconstruction worksheet

Supplied slots	Predict selected result	Observed / interpretation
generation 4 valid; other erased	generation 4	_____
generation 4 valid; other corrupt	generation 4	_____
generations 4 and 5 valid	generation 5	_____
equal generation, unequal bytes	reject	_____
half-range generation difference	reject ambiguous order	_____
both invalid	fail closed, not empty	_____

4 Observe, classify, and bound failures



Alternative text: a graphite pencil timeline shows circle, triangle, and star cards above matching single-card bins. A lowered barrier, an hourglass below the timeline, and then a raised barrier separate those single cards from two later star cards above bins containing stacked copies.

Evidence retains source kind, source ID, configuration revision, occurrence time, sequence, identifier, and complete producer status. Same-domain sequence deltas 1...INT32.MAX move forward. Delta zero is an exact replay only; changed payload at the same sequence is malformed. 0x80000000 is ambiguous, and larger deltas regress.

An unknown identifier increments a saturating failure count once per admitted new sequence. Reaching maximumFailures begins lockout for exactly lockoutDuration; the exact expiry boundary is usable. A known result clears failures only after expiry. Structural, source, time, and image faults take precedence and never increment the count.

Condition	Predicted disposition
identifier matches length and bytes	Known; expose bin and binding revision
canonical but unmatched identifier	Unknown; count one new attempt
exact same-sequence replay	age may change; no new attempt
same sequence with one changed byte	Malformed; preserve accepted history
maximum failures reached	LockedOut
known identity at exact lockout expiry	Known; clear failure count
full table during enrollment	DirectoryFull

5 Timing worksheet

Now	Observed at	Sequence	Prediction / reason
20	20	2	_____
40	40	4	_____
40	40	4 exact replay	_____
91	40	4 exact replay	_____
40	41	5	_____
0	0xffffffff0	next	_____

6 Stage enrollment without claiming persistence

Ordinary observation never enrolls. The transaction keeps the live directory unchanged until an external coordinator reports exact reconciled evidence. The caller supplies a nonzero `instanceEpoch`; initialization rejects zero, and the value must be caller-unique across every concurrent or reconstructed registry lifetime so foreign or old candidates cannot become current. No hidden static counter or object/buffer address supplies that identity.

1. `previewEnrollment()` validates a fresh unknown identity, vacant bin, capacity, lockout state, and base generation.
2. `previewExport()` exposes a bounded view of the complete canonical candidate image. It is retryable and does not commit.
3. A future external coordinator may write the inactive durable slot, synchronize it, reread it, and validate it.
4. `acknowledgeExternalCommit()` validates matching owner, generations, operation, slot, view metadata, checksum, bytes, and durable status, then installs the candidate atomically.

E0 boundary. The compile-only replay supplies matching acknowledgement evidence from caller-owned RAM. It performs no EEPROM, SD, RFID, SPI, keypad, or filesystem operation and proves no retention, power-loss survival, wear endurance, or media atomicity.

Case	Prediction	Observation / interpretation
known identity	enrollment rejected	_____
occupied bin	enrollment rejected	_____
valid preview, then cancel	no live binding added	_____
foreign, stale, or earlier-epoch candidate	reject without mutation	_____
negative or mismatched acknowledgement	preserve retryable candidate; reconciliation required	_____
exact successful acknowledgement	install once, advance generation	_____
byte-identical successful duplicate	idempotent success	_____

7 Replay the eight named frames

Use the Lesson 049 compile-only sketch. Predict before inspecting each volatile result cell. Its flow is acquire, configure, start; then observe, decide, and record. Recording changes only fixture-owned memory.

Frame	Scripted action	Prediction	Result / interpretation
1	observe token A	unknown	_____
2	enroll A at bin 2	commit; generation advances	_____
3	observe A	known bin 2	_____
4	observe token B	unknown; failure count 1	_____
5	exact replay B	no second failure	_____
6	preview then cancel token C	directory unchanged	_____
7	observe B	lockout begins	_____
8	observe C during lockout	locked out	_____

8 Separate acquisition, record result, and safe state

Stage	Predict	Observe and interpret
Acquire	validate config, extents, non-overlap, and both images	status: _____
Observe	admit one complete synthetic identity	sequence/disposition: _____
Reset	clear observation, lockout, and ordinary pending work; retain committed directory	snapshot: _____
Shutdown	publish inactive lifecycle; preserve required reconciliation	status: _____

Initialization proves only that a pure object accepted its configuration and borrowed memory. Reset and shutdown prove only software state. None proves an RFID transaction, keypad level, storage write, inactive pin, safe voltage, or power-removal behavior.

9 Resource record

Gate	Record
<code>sizeof(LocalIdentityRegistry)</code>	measured: 121 bytes; hard ceiling: 128 bytes
complete Mega sketch	flash: 7,026 bytes (limit 16,384); static SRAM: 1,113 bytes (limit 1,280)
caller-owned buffers	two 160-byte images, one 160-byte candidate, and eight bounded live bindings included in whole-sketch SRAM
runtime reserve	Mega SRAM: 8,192 bytes; 7,079 bytes remain after 1,113 static bytes; reserving 512 bytes for stack/ISR leaves 6,567 bytes, exceeding the required 768-byte margin
owned resources	no heap, pins, ADC, timers, interrupts, buses, endpoints, supplies, persistent medium, or moving hardware
host evidence	deterministic, corruption, rollover, lifecycle, sanitizer, and AVR size gates: passed for this host-verified E0 publication

These measured cells are compile and layout evidence for the tested publication. They are not bench evidence.

10 Diagnose fixed-record failures

1. Flip each covered byte without updating its checksum. Predict image or entry rejection and no partial live-table mutation.
2. Change identity padding beyond its declared length. Predict noncanonical-image rejection even when significant bytes match.
3. Supply one valid and one corrupt slot. Predict deterministic recovery from the valid slot; do not call it a durable-media recovery test.
4. Interrupt the modeled handoff with indeterminate acknowledgement. Predict that new work remains latched off until exact reconciliation.
5. Reset or shut down during ordinary pending work. Contrast that result with a reconciliation-required transaction, which must not disappear.

11 Blank E1 reader, keypad, and media acceptance record

Leave every field blank until separately qualified powered and media work is performed and reviewed. E1 authorizes no motor, servo, or moving load.

Required field	Bench record
board revision / tested commit	_____
exact RFID reader and token markings	_____
exact keypad and home/stop specimen markings	_____
exact nonvolatile medium / adapter / revision	_____
primary electrical and media references	_____
supply / logic levels / pin and resource map	_____
instrument / tool version	_____
resource-acquisition and rollback prediction	_____
resource-acquisition and rollback observation	_____
non-Serial test point prediction / observation	_____
safe-state prediction / observation	_____
write / synchronize / reread measurement	_____
power-removal and restart result	_____
interpretation / reviewer / date	_____

12 Claim boundary and evidence links

No retail kit name establishes a reader chip, protocol, voltage, pinout, keypad topology, or storage technology. UID-shaped bytes do not become a credential. Checksums do not become cryptographic integrity. Supplied RAM images do not become persistent media. Physical position and dispensing remain unknown; restrained actuator and gate work belongs to E2.

- Public interface
- Deterministic host tests
- Compile-only synthetic-record replay
- Reviewed design and open E1/E2 gates

The searchable HTML lesson is the hardware-independent accessible route. This untagged print edition makes no PDF/UA, powered-reader, persistent-media, motion, or physical-acceptance claim.